

● EASY

● GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

04

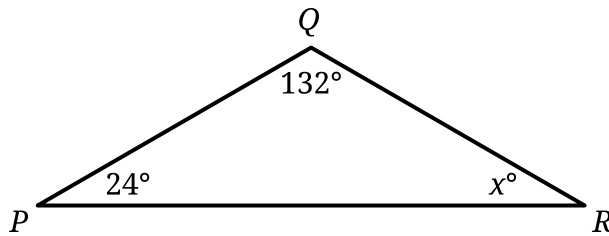
10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY

The width of a rectangle is 7 centimeters. The length of the rectangle is 40 centimeters longer than the width. What is the area, in square centimeters, of this rectangle?

- A. 7
- B. 14
- C. 54
- D. 329

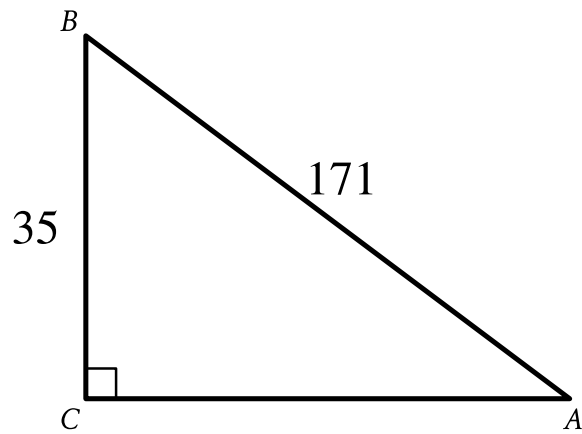


Note: Figure not drawn to scale.

- The measure of angle upper P is 24° .
- The measure of angle upper Q is 132° .
- The measure of angle upper R is x° .
- A note indicates the figure is not drawn to scale.

In the triangle shown, $PQ = QR$. What is the value of x ?

- A. 156
- B. 66
- C. 48
- D. 24



Note: Figure not drawn to scale.

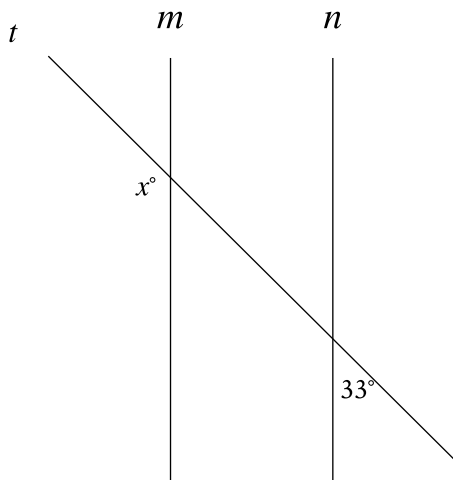
- Angle upper C is a right angle.
- The length of side upper A upper B is 171.
- The length of side upper B upper C is 35.
- A note indicates the figure is not drawn to scale.

In the right triangle shown, what is the value of $\sin A$?

- A. $\frac{1}{171}$
- B. $\frac{35}{171}$
- C. $\frac{171}{35}$
- D. 171

A rectangle has a length of 56 inches and a width of 28 inches. What is the area, in square inches, of the rectangle?

- A. 28
- B. 84
- C. 168
- D. 1,568



Note: Figure not drawn to scale.

- Clockwise from top left, the lines are labeled t , m , and n .
- Line t intersects both line m and line n .
- At the intersection of line t and line m , 1 angle is labeled clockwise from top left as follows:
 - Bottom left: x°
- At the intersection of line t and line n , 1 angle is labeled clockwise from top left as follows:
 - Bottom right: 33°
- A note indicates the figure is not drawn to scale.

In the figure, line m is parallel to line n , and line t intersects both lines. What is the value of x ?

- A. 33
- B. 57
- C. 123

D. 147

Q6

GEOMETRY AND TRIGONOMETRY

A rectangle has an area of 63 square meters and a length of 9 meters. What is the width, in meters, of the rectangle?

A. 7

B. 54

C. 81

D. 567

Q7

GEOMETRY AND TRIGONOMETRY

Triangles EFG and JKL are congruent, where E , F , and G correspond to J , K , and L , respectively. The measure of angle E is 45° and the measure of angle F is 20° . What is the measure of angle J ?

A. 20°

B. 45°

C. 135°

D. 160°

Q8

GRID-IN

GEOMETRY AND TRIGONOMETRY

A right rectangular prism has a length of 11 meters, a width of 8 meters, and a height of 10 meters. What is the volume, in cubic meters, of the prism?

STUDENT-PRODUCED RESPONSE

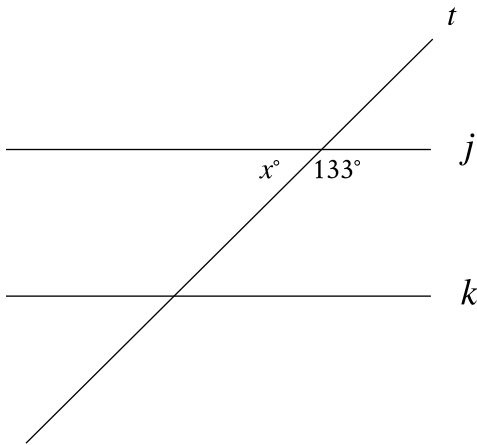
.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GRID-IN

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled t , j , and k .
- Line t intersects both line j and line k .
- At the intersection of line t and line j , 2 angles are labeled clockwise from top left as follows:
 - Bottom right: 133°
 - Bottom left: x°
- A note indicates the figure is not drawn to scale.

In the figure, line j is parallel to line k . What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7

$\frac{\text{ }{\text{ }}}$	$\frac{\text{ }{\text{ }}}$	$\frac{\text{ }{\text{ }}}$	$\frac{\text{ }{\text{ }}}$
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10

GRID-IN

GEOMETRY AND TRIGONOMETRY

What is the area, in square centimeters, of a rectangle with a length of 34 centimeters cm and a width of 29 cm?

STUDENT-PRODUCED RESPONSE

$\square$	$\square$	$\square$	$\square$
.	$\frac{\text{ }{\text{ }}}$	$\frac{\text{ }{\text{ }}}$	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.